

Chapter IV: Exercise 1 - Part 2

How Fast Was Earth's Engine Running?

i This exercise in brief

Big Question

*Did faster spreading rates in the past lead to younger seafloor and thinner lithosphere?
How did crustal architecture evolve through time?*

What You Will Do

You will test a geodynamic hypothesis using multiple datasets from the Palaeo Data Cube. Starting with one time slice, you will:

1. Map seafloor age, crustal thickness, and lithospheric thickness.
2. Compute global averages and distributions.
3. Compare oceanic versus continental regions across variables.
4. Extend your analysis to the entire Phanerozoic and visualize trends.
5. Create an animated GIF of seafloor age evolution through time.

Why It Matters

The age and thickness of the Earth's oceanic crust and lithosphere reflect plate tectonic processes and mantle dynamics. Understanding these changes helps explain supercontinent cycles, ocean basin evolution, and even long-term climate regulation.

Your Tasks

1. **Single Snapshot Analysis:**
 - 1.1. Load seafloor age, crustal thickness, and lithospheric thickness for one time slice using a reusable WCS URL constructor function.
 - 1.2. Compute mean values and visualize maps with custom color ramps.

- 1.3. Plot a histogram of seafloor age distribution.

2. Exploring Relationships:

- 2.1. Combine the three layers into a single `xr.Dataset`.

- 2.2. Scatter plot seafloor age against lithospheric thickness.

- 2.3. Discuss whether the relationship matches theoretical expectations.

3. Deep-Time Trends:

- 3.1. Automate your workflow for all Phanerozoic time slices using `pandas`.

- 3.2. Separate oceanic and continental pixels using the seafloor age mask.

- 3.3. Plot time series of mean seafloor age, crustal thickness, and lithospheric thickness for both domains.

- 3.4. Interpret major trends and link them to supercontinent cycles.

4. Bonus — Animated Visualization:

- 4.1. Render each time slice as an image frame.

- 4.2. Compile frames into an animated GIF using `imageio`.

- 4.3. Experiment with parameters such as time order, frame rate, and layer choice.

Skills You Will Gain

1. Building reusable functions for WCS data access and plotting.
2. Working with multidimensional arrays using `xarray` and combining layers into a `Dataset`.
3. Storing and analyzing time-series results in a `pandas` `DataFrame`.
4. Visualizing maps, histograms, scatter plots, and multi-panel time-series figures.
5. Creating animated GIFs from raster time series.

Points for Discussion

How do crustal and lithospheric thickness trends relate to supercontinent cycles?

Is the relationship between seafloor age and lithospheric thickness what you would expect theoretically?

What assumptions did you make about data interpretation and resolution?

What uncertainties exist in these reconstructions?

Step-by-step instructions

In the first part of this exercise, we focused on palaeogeography, and learned basic tools to:

- Construct a Web Coverage Service (WCS) request through a simple URL using the `requests` library
- Access the data using the `rasterio` library avoiding going through a manual download
- Plot the map using the `matplotlib` library
- Understand the metadata associated with the data we accessed
- Extract simple statistics from a single map (land & ocean areas)
- Generalize this analysis on all available time steps

In this second part, we will follow a similar path, but we will look other maps from the same source. The idea is to have a look at other aspects of the Earth system that informs us about processes taking place in its inner layers.

We will particularly have a look at three layers that are available through the same WCS requests which are (i) seafloor ages, (ii) crustal thickness, and (iii) lithospheric thickness.

As we did before, we will use the `panalesis_atlas` data store on GeoServer, which serves data in world cylindrical equal area (WCEA), using the ESRI:54034 projection. This is done to ensure proper area calculations, as opposed to data in more common CRS such as EPSG:4326, which use degrees units.

1. Single Snapshot Analysis We will first load the three layers for one single age, similar to what we did in the previous part (100 Ma). You have noticed that we used the WCS url request format quite a few times. Instead of repeating the entire code every time, we can define a quick function that we can call with the correct parameters.

```
def construct_wcs_url(layer_name,
                      time,
                      base_url= "https://geoserver.panalesis.org/geoserver/",
                      workspace = "panalesis_atlas",
                      service_type="WCS",
                      service_version="1.0.0",
                      request_type = "GetCoverage",
                      crs = "EPSG:54034",
                      bbox="-20037508.34,-6363885.33,20037508.34,6363885.33",
                      resx = "10000",
                      resy = "10000",
                      format = "GEOTIFF"):

    url = (rf"{base_url}{workspace}/{service_type.lower()}?service={service_type}&version={s
           rf"request={request_type}&"
           fr"coverage={workspace}:{layer_name}&"
           fr"crs={crs}&"
           rf"bbox={bbox}&"
           rf"resx={resx}&"
```

```

rf"resy={resy}&"
rf"time={time}&"
rf"format={format}")

return url

```

The idea is to simplify the next steps where we will handle multiple layers at the same time, for a given age.

→ Notice the function parameters order and syntax, what do you observe ?

We will use this function on our layers of interest, and see if we can plot them easily. For this, we can also create a function to load and plot the maps with a proper color ramp, using the URL, the defined color ramp and a title.

```

import requests
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.colors import ListedColormap, BoundaryNorm
from rasterio.io import MemoryFile

def plot_map(wcs_url,colormap,title):
    data = requests.get(wcs_url).content
    with MemoryFile(data) as memfile:
        with memfile.open() as dataset:
            raster = dataset.read(1).astype(float)
            nodata = dataset.nodata
            if nodata is not None:
                raster = np.ma.masked_equal(raster, nodata)
            else:
                raster = np.ma.masked_invalid(raster)
            bounds = [entry[0] for entry in colormap]
            colors = [entry[1] for entry in colormap[:-1]]
            cmap = ListedColormap(colors)
            norm = BoundaryNorm(
                bounds,
                ncolors=len(colors)
            )
            cmap.set_bad(color='white')
            bounds = dataset.bounds
            extent = [
                bounds.left,

```

```

        bounds.right,
        bounds.bottom,
        bounds.top
    ]

    fig, ax = plt.subplots()
    img = ax.imshow(
        raster,
        cmap=cmap,
        norm=norm,
        extent=extent,
        origin="upper")
    cbar = plt.colorbar(img, ax=ax)
    cbar.set_label(title)
    ax.set_xlabel("Longitude (m)")
    ax.set_ylabel("Latitude (m)")
    plt.show()
    return raster

```

Once these two functions are defined (url constructor + plotter), it becomes very easy to call them. Just define a color ramp of your choosing and call it. Let's focus on seafloor ages:

```

# Seafloor ages
seafloor_ages_url = construct_wcs_url(
    layer_name="seafloor_ages",
    time="2100-01-01T00:00:00.000Z"
)

print(seafloor_ages_url)
sf_colormap_entries = [
    (0, "#30123b", "0"),
    (30, "#4040a2", "30"),
    (60, "#4680f6", "60"),
    (90, "#25c0e7", "90"),
    (120, "#1ad2d2", "120"),
    (150, "#96fe44", "150"),
    (180, "#e9d539", "180"),
    (210, "#fe9029", "210"),
    (240, "#e2430a", "240"),
    (270, "#c52603", "270")
]

```

```
raster = plot_map(
    seafloor_ages_url,
    sf_colormap_entries,
    "Seafloor Age (Myr)"
)
```

Now that we have retrieved the data related to seafloor ages, we are going to take a look at how the ages are distributed. In other words, we are going to investigate if, for our 100 Ma reconstruction, the oceanic crust is rather young, old, intermediate or if the distribution of ages is more complex.

A simple method to do this is to create an histogram that will show the pixel counts for a given number of fixed intervals (bins). Fortunately for us, the **rasterio** library has an in-built function that directly creates an histogram from a raster. we can simply call it like this:

```
import rasterio
from rasterio.plot import show_hist

rasterio.plot.show_hist(
    raster,
    bins=30,
    masked=True,
    title="Seafloor Age Histogram",
    label="100 Ma"
)
```

→ What do you observe on the histogram ? How are the ages distributed ?

→ Given your knowledge about the present-day world, can you guess what the distribution will be if we do a query at 0Ma ? Try it !

Another very quick insights we can get is to calculate the mean value for a given age, by simply calling:

```
mean_age = raster.mean()
print(f"Mean seafloor age is {mean_age}")
```

We are now going to check this for the entire Phanerozoic, as we did for the land and ocean areas. Again, we will simply create a loop that will calculate the mean value of every available raster. Let us use the same function as before, with an update for default parameters that are (for our demo) not going to change

```

from xml.etree import ElementTree as ET

def get_available_times(
    layer_name,
    base_url="https://geoserver.panalexis.org/geoserver/",
    workspace= "panalexis_atlas",
    service_type = "WCS",
    service_version="1.0.0"
):
    describe_url = (
        f"{base_url}{workspace}/{service_type.lower()}"
        f"?service={service_type}&version={service_version}"
        f"&request=DescribeCoverage&coverage={workspace}:{layer_name}"
    )

    response = requests.get(describe_url)
    root = ET.fromstring(response.content)

    times = sorted(set(el.text for el in root.iter() if el.tag.endswith('timePosition')) and
    return sorted(times)

```

Which we can now use to retrieve all available time steps, and retrieve each map, calculate its mean age value, and finally plot a timeseries chart of the mean age evolution:

```

layer_name = "seafloor_ages"
times = get_available_times(
    layer_name=layer_name
)

results = {}
for time in times:
    geological_age = int(time[:4]) - 2000
    wcs_url = construct_wcs_url(
        layer_name=layer_name,
        time=time
    )
    data = requests.get(wcs_url).content
    with MemoryFile(data) as memfile:
        with memfile.open() as dataset:
            raster = dataset.read(1).astype(float)
            nodata = dataset.nodata
            if nodata is not None:

```

```

        raster = np.ma.masked_equal(
            raster,
            nodata
        )
    else:
        raster = np.ma.masked_invalid(raster)
    mean_age = raster.mean()
    results[geological_age] = {'sf_mean_age': float(mean_age)}
    del data, raster

ages = list(results.keys())
seafloor_ages = [results[age]['sf_mean_age'] for age in ages]

plt.figure(figsize=(4, 3.5))

plt.plot(
    ages,
    seafloor_ages,
    color='red',
    marker='o',
    markersize=4
)
plt.xlabel('Age (Ma)')
plt.ylabel('Mean Seafloor Age (Myr)')
plt.gca().invert_xaxis()
plt.gca().set_box_aspect(1)

plt.tight_layout()
plt.show()

```

→ What do you observe ? Can you guess the reason for these variations ?

In the last two time-series we plotted (land/ocean area , seafloor ages), we only considered one input map at a time with the **rasterio** library. This is good, but what if we want to “pack” different sources together and derive insights from several layers ?

For this, we are going to use **xarray**, a Python library that deals with multidimensional arrays. An array is a data structure that stores a collection of elements of the same type, organized in a grid-like structure where each element can be retrieved using its position. In our case, as our maps share the same resolution and extent, the elements inside our array are the map pixels.

Organizing our data in arrays will prove very useful for combined analysis of several variables. Below is a function that creates an array from a WCS URL:


```

import xarray as xr
import rioxarray

def load_raster_as_dataarray(layer_name, time, name):
    url = construct_wcs_url(
        layer_name=layer_name,
        time=time
    )
    data = requests.get(url).content
    with MemoryFile(data) as memfile:
        with memfile.open() as dataset:
            raster = dataset.read(1).astype(float)
            nodata = dataset.nodata
            if nodata is not None:
                raster[raster == nodata] = np.nan
            raster[~np.isfinite(raster)] = np.nan
            da = xr.DataArray(
                raster,
                dims=["y", "x"],
                name=name
            )
    return da

```

This function opens a raster from a WCS URL as we did before, but then creates an `xr.DataArray` class object to create a multidimensional array: two spatial dimensions (x and y, according to the pixel index) describing the position (index) of each pixel.

We can then create three data arrays (one for each layer), for a given time:

```

time = "2100-01-01T00:00:00.000Z"

sf_age = load_raster_as_dataarray(
    "seafloor_ages",
    time,
    "seafloor_age"
)
crustal = load_raster_as_dataarray(
    "crustal_thickness",
    time,
    "crustal_thickness"
)
litho = load_raster_as_dataarray(

```

```

    "lithospheric_thickness",
    time,
    "lithospheric_thickness"
)

```

As they share the same dimensions, these 3 arrays can be combined in a single `xr.Dataset`:

```

ds = xr.Dataset({
    "seafloor_age": sf_age,
    "crustal_thickness": crustal,
    "lithospheric_thickness": litho
})

```

The idea behind this construction is simply to get a dataset that has consistent coordinates and various pixel properties. The structure of our dataset will look something like:

```

Dataset
  seafloor_age          (y: 1273, x: 4018)
  crustal_thickness     (y: 1273, x: 4018)
  lithospheric_thickness (y: 1273, x: 4018)
coordinates: x, y (shared)

```

Taking advantage of this simple structure, it is now easier to have a glance at all variables with one simple call:

```

for var in ds.data_vars:
    print(f"\n{var}:")
    print(f"  mean : {float(ds[var].mean()):.2f}")
    print(f"  std  : {float(ds[var].std()):.2f}")
    print(f"  min  : {float(ds[var].min()):.2f}")
    print(f"  max  : {float(ds[var].max()):.2f}")

```

Now, we will have a look at the relationship between the seafloor age and the thickness of the lithosphere. Let's do a simple plot that will show this relationship. We first need to do some pre-filtering of our data. The code below will (i) “flatten” our multidimensional array into a single dimensional one, (ii) discard non oceanic pixels, and (iii) discard any invalid pixels:

```

age_flat = ds["seafloor_age"].values.flatten()
litho_flat = ds["lithospheric_thickness"].values.flatten()

ocean_mask = np.isfinite(age_flat) & (age_flat > 0)
age_valid = age_flat[ocean_mask]
litho_valid = litho_flat[ocean_mask]

both_valid = np.isfinite(age_valid) & np.isfinite(litho_valid)
age_valid = age_valid[both_valid]
litho_valid = litho_valid[both_valid]

```

Once this filtering is done, we can plot our two variables:

```

plt.figure(figsize=(5, 5))
plt.scatter(
    age_valid,
    litho_valid,
    s=0.5,
    alpha=0.3,
    color="steelblue"
)
plt.xlabel("Seafloor Age (Myr)")
plt.ylabel("Lithospheric Thickness (km)")
plt.show()
del litho, crustal, sf_age, age_flat, litho_flat, ocean_mask, age_valid, litho_valid

```

→ What do you observe ? Is the relationship coherent with what you would expect from your knowledge ?

→ Can you find a mathematical function that describes this trend? Try using `numpy.polyfit` to fit a curve to the data.

Now that we understand the relationship between seafloor age and oceanic lithosphere, we will extend our time-series visualization to continental crust and lithosphere thickness. As we will plot the entire Phanerozoic evolution, our previous approach to use `xr.DataArray` (single map) or `xr.Dataset` (multiple maps) for a single time-step is now limited. For this, we will use the very well known and widely used `pandas` Python library.

The structure of the code below should be now well known to you. We simply iterate through the time-steps listed in `times`. For each step, we load the seafloor age, crustal & lithospheric thickness as `xr.DataArray`, which we then combine into a `xr.Dataset`. We then discriminate between oceanic versus continental regions, based on whether or not the seafloor age raster value is valid or not.

The new aspect here is how we store the results. At the beginning of the code, you can see we create an empty list to store results, which we populate with the mean values for each time. After the loop, we create a `pd.DataFrame` containing all mean values from all ages.

```
import pandas as pd
times = [
    '2006-01-01T00:00:00.000Z', '2011-01-01T00:00:00.000Z',
    '2015-01-01T00:00:00.000Z', '2020-01-01T00:00:00.000Z',
    '2033-01-01T00:00:00.000Z', '2040-01-01T00:00:00.000Z',
    '2048-01-01T00:00:00.000Z', '2056-01-01T00:00:00.000Z',
    '2068-01-01T00:00:00.000Z', '2084-01-01T00:00:00.000Z',
    '2094-01-01T00:00:00.000Z', '2100-01-01T00:00:00.000Z',
    '2113-01-01T00:00:00.000Z', '2120-01-01T00:00:00.000Z',
    '2133-01-01T00:00:00.000Z', '2140-01-01T00:00:00.000Z',
    '2154-01-01T00:00:00.000Z', '2165-01-01T00:00:00.000Z',
    '2180-01-01T00:00:00.000Z', '2200-01-01T00:00:00.000Z',
    '2210-01-01T00:00:00.000Z', '2220-01-01T00:00:00.000Z',
    '2230-01-01T00:00:00.000Z', '2240-01-01T00:00:00.000Z',
    '2250-01-01T00:00:00.000Z', '2270-01-01T00:00:00.000Z',
    '2290-01-01T00:00:00.000Z', '2300-01-01T00:00:00.000Z',
    '2315-01-01T00:00:00.000Z', '2331-01-01T00:00:00.000Z',
    '2350-01-01T00:00:00.000Z', '2370-01-01T00:00:00.000Z',
    '2383-01-01T00:00:00.000Z', '2393-01-01T00:00:00.000Z',
    '2408-01-01T00:00:00.000Z', '2420-01-01T00:00:00.000Z',
    '2444-01-01T00:00:00.000Z', '2463-01-01T00:00:00.000Z',
    '2475-01-01T00:00:00.000Z', '2489-01-01T00:00:00.000Z',
    '2500-01-01T00:00:00.000Z', '2518-01-01T00:00:00.000Z',
    '2535-01-01T00:00:00.000Z', '2545-01-01T00:00:00.000Z'
]
results = []

for time in times:
    geological_age = int(time[:4]) - 2000
    sf_age = load_raster_as_dataarray(
        "seafloor_ages",
        time,
        "seafloor_age"
    )
    crustal = load_raster_as_dataarray(
        "crustal_thickness",
        time,
        "crustal_thickness"
    )
```

```

litho = load_raster_as_dataarray(
    "lithospheric_thickness",
    time,
    "lithospheric_thickness"
)

ds = xr.Dataset({
    "seafloor_age": sf_age,
    "crustal_thickness": crustal,
    "lithospheric_thickness": litho
})

age_vals = ds["seafloor_age"].values
crustal_vals = ds["crustal_thickness"].values
litho_vals = ds["lithospheric_thickness"].values

ocean_mask = np.isfinite(age_vals)
continental_mask_crustal = ~ocean_mask & np.isfinite(crustal_vals)
continental_mask_litho = ~ocean_mask & np.isfinite(litho_vals)

results.append({
    "age": geological_age,
    "mean_seafloor_age": float(np.nanmean(age_vals[ocean_mask])),
    "mean_crustal_ocean": float(np.nanmean(crustal_vals[ocean_mask])),
    "mean_crustal_continental": float(np.nanmean(crustal_vals[continental_mask_crustal])),
    "mean_litho_ocean": float(np.nanmean(litho_vals[ocean_mask])),
    "mean_litho_continental": float(np.nanmean(litho_vals[continental_mask_litho]))
})

del ds, sf_age, crustal, litho

df = pd.DataFrame(results).sort_values("age").reset_index(drop=True)

```

Creating a `pd.DataFrame` allows us to pack our data in a tabular format: columns with names and rows (indices). This organization is used in all data science fields. Once we have our dataframe, it is very straightforward to plot the various time series.

For instance, the code below creates a figure with two rows and two columns. We can then arrange each subplot as we prefer. Notice how we simply access subsets of our dataframe to display them in each subplot.

```

fig, axes = plt.subplots(
    nrows=2,
    ncols=2,
    figsize=(8, 8)
)

# Top-left: Mean Seafloor Age
axes[0, 0].plot(
    df["age"],
    df["mean_seafloor_age"],
    color="steelblue",
    marker="o",
    markersize=3
)
axes[0, 0].set_ylabel("Mean Seafloor Age (Myr)")

# Top-right: Crustal Thickness (both)
axes[0, 1].plot(
    df["age"],
    df["mean_crustal_continental"],
    color="sienna",
    marker="o",
    markersize=3
)
axes[0, 1].set_ylabel("Continental Crust \n Mean Thickness (km)")

# Bottom-left: Lithospheric Thickness - Ocean
axes[1, 0].plot(
    df["age"],
    df["mean_litho_ocean"],
    color="steelblue",
    marker="o",
    markersize=3
)
axes[1, 0].set_ylabel("Oceanic Lithosphere \n Mean Thickness (km)")
axes[1, 0].set_xlabel("Age (Ma)")

# Bottom-right: Lithospheric Thickness - Continental
axes[1, 1].plot(
    df["age"],
    df["mean_litho_continental"],
    color="sienna",

```

```

        marker="o",
        markersize=3
    )
    axes[1, 1].set_ylabel("Continental Lithosphere \n Mean Thickness (km)")
    axes[1, 1].set_xlabel("Age (Ma)")

    for ax in axes.flat:
        ax.axvspan(
            320, 175,
            alpha=0.15,
            color="grey",
            label="Pangea"
        )
        ax.invert_xaxis()

plt.tight_layout()
plt.show()

```

→ Are there any interesting trends ? Do you notice some odd results ?

→ Using your knowledge about the supercontinent cycles, and the timing for formation/break-up of these supercontinents, do you see these cycles linked with the data we have ?

Bonus Finally, we will learn how to create an animated GIF to visualize one layer evolution through the Phanerozoic. For this we will need another two libraries called `io` and `imageio`, designed to deal with files management, and images and video files, respectively.

```

import io
import imageio

def render_frame(wcs_url, colormap, title, geological_age):
    data = requests.get(wcs_url).content
    with MemoryFile(data) as memfile:
        with memfile.open() as dataset:
            raster = dataset.read(1).astype(float)
            nodata = dataset.nodata
            if nodata is not None:
                raster = np.ma.masked_equal(
                    raster,
                    nodata
                )
            else:
                raster = np.ma.masked_invalid(raster)

```

```

        bounds = [entry[0] for entry in colormap]
        colors = [entry[1] for entry in colormap[:-1]]
        cmap = ListedColormap(colors)
        norm = BoundaryNorm(
            bounds,
            ncolors=len(colors)
        )
        cmap.set_bad(color='white')
        bounds = dataset.bounds
        extent = [
            bounds.left,
            bounds.right,
            bounds.bottom,
            bounds.top
        ]
        fig, ax = plt.subplots()
        img = ax.imshow(
            raster,
            cmap=cmap,
            norm=norm,
            extent=extent,
            origin="upper"
        )
        cbar = plt.colorbar(img, ax=ax)
        cbar.set_label(title)
        ax.set_title(f"{geological_age} Ma")
        ax.set_xlabel("Longitude (m)")
        ax.set_ylabel("Latitude (m)")
        buf = io.BytesIO()
        plt.savefig(
            buf,
            format="png",
            dpi=100
        )
        plt.close(fig)
        buf.seek(0)
        return imageio.imread(buf)

frames = []

for time in times:
    geological_age = int(time[:4]) - 2000

```



```

wcs_url = construct_wcs_url(
    layer_name="seafloor_ages",
    time=time
)
frame = render_frame(
    wcs_url,
    sf_colormap_entries,
    "Seafloor Age (Myr)",
    geological_age
)
frames.append(frame)

output_gif = "seafloor_ages_phanerozoic.gif"
imageio.mimsave(
    output_gif,
    frames,
    fps=4,
    loop=0
)
print(f"GIF saved as: {output_gif}")

```

→ Play around with the code above and try to change some parameters (revert the time to have something going from the oldest age to present, adjust the duration of each map display, change the layer and symbology).

💡 Links to useful libraries and tools

Python libraries

- [xarray](#) A Python library for working with labeled multi-dimensional arrays, providing a Dataset structure that combines multiple variables sharing the same coordinate system.
- [rioxarray](#) An extension of xarray that adds geospatial capabilities, enabling CRS-aware operations on raster datasets directly within the xarray framework.
- [pandas](#) A Python library for data manipulation and analysis, providing a DataFrame structure that organizes tabular data with named columns and row indices.
- [io](#) A Python standard library module for handling Input/Output (I/O) operations, such as reading data in or writing data out.
- [imageio](#) A Python library for reading and writing a wide range of image and video formats, offering a simple interface to load image data as NumPy arrays.